

NCERT Solutions For Class 12 Computer Science

Chapter 10 – Computer Networks

1. Expand the following:

(a) ARPANET

(b) MAC

(c) ISP

(d) URI

Answer:

(a) **ARPANET**: Advanced Research Project Agency Network

(b) **MAC**: Media Access Control

(c) **ISP**: Internet Service Provider

(d) **URI**: Uniform Resource Identifier

2. What do you understand by the term network?

A **network** is an interconnected collection of autonomous computing devices that can share and exchange information and resources. In computer context, it refers to connecting two or more computers/devices so they can communicate and share data, hardware, and software resources.

3. Mention any two main advantages of using a network of computing devices.

Resource Sharing: Computer networks allow sharing of data, files, software, and hardware like printers and storage across connected devices, regardless of physical location.

Reliability: Networked devices can have backup copies on other computers, so if one device is unavailable, the information can be accessed from another, improving reliability.

4. Differentiate between LAN and WAN.

LAN	WAN
Stands for Local Area Network.	Stands for Wide Area Network.
Covers a small area (room to campus, up to 1 km).	Covers a large area (city, country, or world).
High speed and more secure.	Slower speed due to large area, less secure comparatively.
Easy and less costly to set up.	Complex and more expensive to set up and maintain.

5. Write down the names of few commonly used networking devices.

Answer:

- (a) Modem
- (b) Hub
- (c) Switch
- (d) Router
- (g) ateway

6. Two universities in different States want to transfer information. Which type of network do they need?

They need a **Wide Area Network (WAN)**, as WANs connect devices and LANs over long geographical distances, such as across states or countries.

7. Define the term topology. What are the popular network topologies?

Topology refers to the arrangement of computers and peripherals in a network.

Popular network topologies include Mesh, Ring, Bus, Star, and Tree (Hybrid) topologies.

8. How is tree topology different from bus topology?

Tree topology is a hierarchical structure combining various basic

topologies like star, ring, or bus. Each branch can host more nodes. Data passes through a central device to branches.

In **bus topology**, all devices are connected to a single central backbone (bus) and data travels along this medium in both directions.

9. Identify the type of topology from the following:

- (a) Each node is connected with the help of a single cable.
- (b) Each node is connected with central switching through independent cables.

Answer:

- (a) Bus Topology
- (b) Star Topology

10. What do you mean by a modem? Why is it used?

A **modem** is an electronic device (MODulator-DEModulator) that converts digital data from computers into analog signals and vice versa. It is used for transmission over telephone or communication lines, enabling internet connectivity.

11. Explain the following devices:

- (a) Switch
- (b) Repeater
- (c) Router
- (d) Gateway

(e) NIC

(a) Switch: A switch is a networking device that connects multiple computers or devices in a LAN and selectively forwards data packets to the correct destination. It is more efficient than a hub and does not forward corrupted signals.

(b) Repeater: A repeater regenerates and amplifies weak or degraded signals so that data can travel longer distances without error.

(c) Router: A router connects multiple networks and routes data intelligently between them. Routers also connect LANs to the internet and can handle various protocols.

(d) Gateway: A gateway is the entry/exit point connecting an internal network to external networks, often converting protocols and also integrating security features.

(e) NIC: The Network Interface Card (NIC) allows computers to connect to a network physically. Each NIC has a unique MAC address for identification.

12. Draw a network layout of star topology and bus topology connecting five computers.

Answer:

Star Topology:

Each computer is connected independently to a central hub/switch.

Bus Topology:

All five computers are connected to a single central backbone (bus) through drop lines.

Star Topology



Bus Topology



13. What is the significance of MAC address?

A **MAC address** is a unique physical identifier assigned to the network interface card (NIC) of a device. It helps identify each device on a network at the hardware level and is essential for managing network communication and security.

14. How is IP address different from MAC address? Discuss briefly.

An **IP address** is a logical (changeable) address used to uniquely identify

and communicate with a device on a network; it can change across networks. A **MAC address** is a permanent, physical address assigned to the hardware (NIC) that does not change and identifies the device itself.

15. What is DNS? What is a DNS server?

DNS (Domain Name System) is a system that translates human-readable domain names (like www.ncert.nic.in) into machine-readable IP addresses. A **DNS server** is a special server that performs this translation, enabling users to access websites using domain names instead of IP numbers.

16. Sahil, a class X student, has just started understanding the basics of Internet and web technologies. He is a bit confused between the terms “World Wide Web” and “Internet”. Help him in understanding both terms with suitable examples of each.

The **Internet** is the global network of interconnected computers that allows devices worldwide to communicate. The **World Wide Web (WWW)** is a service on the Internet consisting of interlinked web pages accessed via browsers using URLs (like www.ncert.nic.in).

The Internet provides the infrastructure, while WWW is one of the services using that infrastructure.